Probability and Statistics

Random Variables

Dr. Yasir Ali (yali@ceme.nust.edu.pk)

DBS&H, CEME-NUST

May 10, 2022

Types of Sample spaces

Discrete Sample Spaces

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space.**

A random variable is called a d**iscrete random variable** if its set of possible outcomes is countable.

Continuous Sample Spaces

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

A random variable can take on values on a continuous scale, it is called a **continuous random variable**.

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. Let Y be the number of red balls. The possible outcomes and the values y of the random variable Y are

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. Let Y be the number of red balls. The possible outcomes and the values y of the random variable Y are

Sample Space	y
RR	2
RB	1
BR	1
BB	0

Suppose a sampling plan involves sampling items from a process until a defective is observed. The evaluation of the process will depend on how many consecutive items are observed. In that regard, let X be a random variable defined by the number of items observed until a defective is found.

With N a non-defective and D a defective, sample spaces are $S = \{D\}$ given $X = 1$, $S = \{ND\}$ given $X = 2$, $S = \{NND\}$ given $X = 3$, and so on.

- Let X be the random variable defined by the waiting time, in hours, between successive speeders spotted by a radar unit. The random variable X takes on all values x for which $x \geq 0$.

1. The number of cats in a shelter at any given time.
2. The weight of newborn babies.
3. The weight of a book in the library.
4. The number of books in the library.
5. The number of days someone lives.
6. The length of time someone lives.

The number of students using the Math Lab per day is found in the distribution below.

X	6	8	10	12	14
P(X)	0.15	0.3	0.35	0.1	0.1

What is the probability that fewer than 8 or more than 12 use the lab in a given day?

④ The x less than 8 is 6, the X more than 12 is 14.
Hence the answer $P = P(6) + P(14) = 0.25$

Random Variable

A random variable is a variable whose values are determined by chance.

A **variable** was defined as a characteristic or attribute that can assume different values. Various letters of the alphabet, such as **X, Y**, or **Z**, are used to represent variables

For example, if a die is rolled, a letter such as X can be used to represent the outcomes. Then the value that X can assume is 1, 2, 3, 4, 5, or 6, corresponding to the outcomes of rolling a single die.

If two coins are tossed, a letter, say **Y**, can be used to represent ***the number of heads***, in this case 0, 1, or 2.

As another example, if the temperature at 8:00 A.M. is 43° and at noon it is 53°, then the values T that the temperature assumes are said to be random, since they are due to various atmospheric conditions at the time the temperature was taken.

The quantity that we observe in an experiment will be denoted by X and called a random variable (or stochastic variable) because the value it will assume in the next trial depends on chance, on *randomness*.

Types of Random Variables

1. Discrete variables have a finite number of possible values or an infinite number of values that can be counted.
2. Variables that can assume all values in the interval between any two given values are called continuous variables.

If we count (cars on a road, defective screws in a production, tosses until a die shows the first Six), we have a discrete random variable. If we measure (electric voltage, rainfall, hardness of steel), we have a continuous random variable.

Examples of Random Variables(RV)

1. Three coins are tossed and if X is the random variable for the number of heads, then X assumes the value **0, 1, 2, or 3**.
2. Let X denotes the number of games played in a series between two teams where a team to win four games wins the the Series. Then X assumes the value **4,5,6 or 7**.
3. A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. Let X denotes the number of defective laptops then X assumes the value **0, 1, 2, or 3**. If a school makes a random purchase of 2 of these computers, X denotes the number of defective laptops for the school then X assumes the value **0, 1 or 2**.
4. Let X be a random variable giving the number of heads minus the number of tails in three tosses of a coin then X assumes the value **3, 1,-1 or -3**.

A stockroom clerk hands over three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
BSJ	0
JBS	0

Probability Distribution or Distribution

A probability distribution or, briefly, a distribution, shows

the probabilities of events in an experiment.

A discrete probability distribution consists of

the **values** a random variable can assume and the **corresponding probabilities of the values**.

The probabilities are determined theoretically or by observation.

When three coins are tossed, the sample space is represented as

$$TTT,\ TTH,\ THT,\ HTT,\ HHT,\ HTH,\ THH,\ HHH;$$

If X is the random variable for the number of heads, then X assumes the value 0, 1, 2, or 3.

If X is the random variable for the number of heads, then X assumes the value 0, 1, 2, or 3.

No heads	One head			Two heads			Three heads
TTT	TTH	THT	HTT	HHT	HTH	THH	HHH
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{8}$	$\frac{3}{8}$			$\frac{3}{8}$			$\frac{1}{8}$

Hence, the probability of getting no heads is $\frac{1}{8}$, one head is $\frac{3}{8}$, two heads is $\frac{3}{8}$, and three heads is $\frac{1}{8}$.

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Two Requirements for a Probability Distribution

1. The probability of each event in the sample space must be between or equal to 0 and 1. That is,

$$0 \leq P(X) \leq 1$$

2. The sum of the probabilities of all the events in the sample space must equal 1; that is,

$$\sum P(X) = 1$$

Determine whether each distribution is a probability distribution.

a.

X	0	5	10	15	20
$P(X)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

b.

X	0	2	4	6
$P(X)$	-1.0	1.5	0.3	0.2

c.

X	1	2	3	4
$P(X)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{9}{16}$

d.

X	2	3	7
$P(X)$	0.5	0.3	0.4

Solution

a. Yes, it is a probability distribution.

b. No, it is not a probability distribution,

c. Yes, it is a probability distribution.

d. No, it is not, since $\Sigma P(X) = 1.2$.

Construct a probability distribution for rolling a single die.

Since the sample space is 1, 2, 3, 4, 5, 6 and each outcome has a probability of , the distribution is as shown.

Outcomes X	1	2	3	4	5	6
P(X)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

The baseball World Series is played by the winner of the National League and the American League. The first team to win four games wins the World Series. In other words, the series will consist of four to seven games, depending on the individual victories. The data shown consist of the number of games played in the World Series from 1965 through 2005. (There was no World Series in 1994.) The number of games played is represented by the variable X. Find the probability P(X) for each X, construct a probability distribution, and draw a graph for the data.

X	Number of games played
4	8
5	7
6	9
7	16

X	4	5	6	7
P(X)	0.2	0.175	0.225	0.4

we write $f(x) = P(X = x)$; that is, $f(4) = P(X = 4)$, $f(5) = P(X = 5)$ and so on. .

By definition, a random variable X and its distribution are ***discrete*** if X assumes only ***finitely many*** or ***at most countably many values***

$$x_1, \quad x_2, \quad x_3, \quad \cdots,$$

called ***the possible values of***, with positive probabilities

$$p_1 = P(X = x_1), \quad p_2 = P(X = x_2), \cdots, p_j = P(X = x_j) \cdots,$$

whereas the probability $P(X \in I)$ is zero for any interval I containing no possible value.
Clearly, the discrete distribution of X is also determined by the probability function of X, defined by

$$f(x) = \begin{cases} p_j, & \text{if } x = x_j; \\ 0, & \text{otherwise.} \end{cases} \qquad (j = 1, 2, 3, \cdots), \tag{1}$$

A stockroom clerk hands over three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
BSJ	0
JBS	0

m	0	1	3
p(m)	2/6	3/6	1/6

The set of ordered pairs $(x, f(x))$ is a **probability mass function** of the discrete random variable X if, for each possible outcome x,

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $P(X = x) = f(x)$

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X, defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in \mathbf{R}$.
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_a^b f(x)dx$.

In case of continuous random variables we shall concern with computing probabilities for various intervals such as $P(a < X \leq b)$, $P(W \geq c)$, and so forth. Note that when X is continuous,

$$P(a < X \leq b) = P(a < X < b) + P(X = b) = P(a < X < b).$$

1. Suppose that the error in the reaction temperature, in ^{o}C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & otherwise \end{cases}$$

(a) Obviously, $f(x) \geq 0$. To verify condition 2 in Definition 3.6, we have

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-1}^{2} \frac{x^2}{3} dx = \frac{x^3}{9}\Big|_{-1}^{2} = \frac{8}{9} + \frac{1}{9} = 1.$$

(b) Using formula 3 in Definition 3.6, we obtain

$$P(0 < X \leq 1) = \int_0^1 \frac{x^2}{3} dx = \frac{x^3}{9}\Big|_0^1 = \frac{1}{9}.$$

   (a) Verify that f(x) is a density function.
   (b) Find $P(0 < X \leq 1)$.

2. Find the constant c such that the function

$$f(x) = \begin{cases} cx^2 & 0 < x < 3 \\ 0 & otherwise \end{cases}$$

is a density function.

Ans: 1/9

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and 2. Now find $f(x)$, for $x=0,1,2$

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and 2. Now find f(x), for x=0,1,2

Number of ways of selecting x defective computers are

$$n(x) = C_x^3 C_{2-x}^{17}$$

$$n(0) = C_0^3 C_2^{17}$$
$$n(1) = C_1^3 C_1^{17}$$
$$n(2) = C_2^3 C_0^{17}$$

Number of ways of selecting 2 computers are $\boxed{C_2^{20}}$

x	0	1	2
f(x)	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

Find the probability distribution of boys and girls in families with 3 children, assuming equal probabilities for boys and girls.

In case of n mutually independent trials, where each trial had just two possible outcomes, A and A^c, with respective probabilities p and $q = 1 - p$. It was found that the probability of getting exactly $xA's$ in the n trials is

$$C^n_x p^x q^{n-x}.$$

Under the assumption that successive births (the "trial") are independent as far as the sex of the child is concerned. Thus, with A being the event "a boy", $n = 3$, $p = \frac{1}{2} = q$, we have

$$P(\text{Exactly } x \text{ boys}) = P(X = x) = f(x) = C^3_x p^x q^{3-x}$$

Find the probability distribution of boys and girls in families with 3 children, assuming equal probabilities for boys and girls.

In case of n mutually independent trials, where each trial had just two possible outcomes, A and A^c, with respective probabilities p and $q = 1 - p$. It was found that the probability of getting exactly $xA's$ in the n trials is

$$C_x^n p^x q^{n-x}.$$

Under the assumption that successive births (the "trial") are independent as far as the sex of the child is concerned. Thus, with A being the event "a boy", $n = 3$, $p = \frac{1}{2} = q$, we have

$$P(\text{Exactly } x \text{ boys}) = P(X = x) = f(x) = C_x^3 p^x q^{3-x}$$

x	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

PMF & PDF

The set of ordered pairs $(x, f(x))$ is called the **Probability Mass Function(PMF)** of the discrete random variable X.

X	4	5	6	7
P(X)	0.2	0.175	0.225	0.4

x	0	1	2
f(x)	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

x	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

Probability Density Function (PDF)

The function f(x) is a probability density function (PDF) for the continuous random variable X.

Cumulative Distribution Function (CDF)

From PMF, we obtain the **cumulative distribution function** by taking sums,

$$F(x) = \sum_{x_j \leq x} f(x_j) = \sum_{x_j \leq x} p_j. \tag{2}$$

The **cumulative distribution function F(x)** of a continuous random variable X with density function f(x) is,

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt \tag{3}$$

We can write $P(a < X < b) = F(b) - F(a)$ and $f(x) = \frac{dF(x)}{dx}$

If X takes on only a finite number of values $x_1, x_2, \cdots, x_n$, then the distribution function is given by

$$F(x) = \begin{cases} 0 & -\infty \leq x \leq x_1 \\ f(x_1) & x_1 \leq x \leq x_2 \\ f(x_1) + f(x_2) & x_2 \leq x \leq x_3 \\ f(x_1) + f(x_2) + f(x_3) & x_3 \leq x \leq x_4 \\ \vdots & \vdots \\ f(x_1) + f(x_2) + \cdots + f(x_n) & x_n \leq x \leq \infty \end{cases}$$

A stockroom clerk hands over three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
BSJ	0
JBS	0

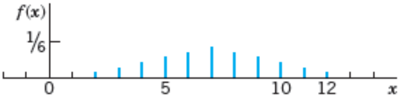
m	0	1	3
p(m)	2/6	3/6	1/6

$$F(m) = \begin{cases} 0 & m < 0 \\ 2/6 & 0 \leq m < 1 \\ 5/6 & 1 \leq m < 3 \\ 1 & m > 3 \end{cases}$$

The random variable $X =$ **Sum of the two numbers two fair dice turn up** is discrete and has the possible values $(1+1) = 2, (1+2) = 3, \cdots, (6+6) = 12$

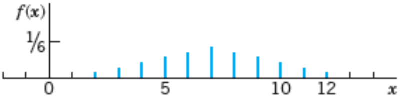
The random variable X = **Sum of the two numbers two fair dice turn up** is discrete and has the possible values $(1+1) = 2, (1+2) = 3, \cdots, (6+6) = 12$

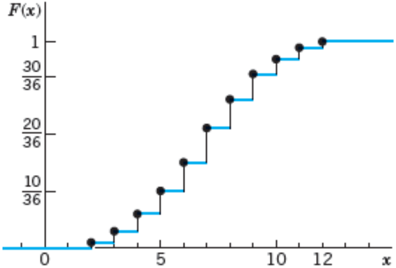
x	2	3	4	5	6	7	8	9	10	11	12
$f(x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36
$F(x)$	1/36	3/36	6/36	10/36	15/36	21/36	26/36	30/36	33/36	35/36	36/36



The random variable X = **Sum of the two numbers two fair dice turn up** is discrete and has the possible values $(1+1)=2, (1+2)=3, \cdots, (6+6)=12$

x	2	3	4	5	6	7	8	9	10	11	12
$f(x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36
$F(x)$	1/36	3/36	6/36	10/36	15/36	21/36	26/36	30/36	33/36	35/36	36/36





Probability function $f(x)$ and distribution function F(x) of the random variable X = Sum of the two numbers obtained in tossing two fair dice once

For the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & otherwise \end{cases}$$

find F(x), and use it to evaluate $P(0 < X \leq 1)$.

For the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & otherwise \end{cases}$$

find F(x), and use it to evaluate $P(0 < X \leq 1)$.

$$\boxed{F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt}$$

$$\boxed{P(a < X < b) = F(b) - F(a)}$$

$$\begin{aligned} F(x) =& P(X \leq x) = \int_{-\infty}^{x} f(t)dt \\ =& \int_{-\infty}^{-1} f(t)dt + \int_{-1}^{x} f(t)dt \end{aligned}$$

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x^3+1}{9} & -1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

$$P(0 < X \leq 1) = F(1) - F(0)$$

The distribution function for a random variable X is

$$F(x) = \begin{cases} 1 - e^{-2x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Find **(a) the density function**, **(b) the probability that** $X > 2$ and **(c) the probability that** $-3 < X \leq 4$.

$$\boxed{f(x) = \frac{d}{dx} F(x)} \quad \boxed{P(X > 2) = \int_2^{\infty} f(x)dx}$$

$$P(-3 < X \leq 4) = \int_{-3}^{4} f(x)dx$$

Can you calculate these using $F(x)$?

Find the probability distribution for the number of jazz CDs when 4 CDs are selected at random from a collection consisting of 5 jazz CDs, 2 classical CDs, and 3 rock CDs. Express your results by means of a formula.

Let X be a random variable representing the number of jazz CD's among the 4 selected CDs. Then X assumes the values 0,1,2,3,4

- Total ways of selection of 4 CDs out of 10
- Ways of selecting x jazz CDs

Let W be a random variable giving the number of heads minus the number of tails in three tosses of a coin. List the elements of the sample space S for the three tosses of the coin and to each sample point assigna value w of W.

1. Find the probability distribution of the random variable W assuming that the coin is biased so that a head is twice as likely to occur as a tail.
2. Find the cumulative distribution function of the random variable W. Using $F(w)$, find
   (a) $P(W > 0)$;
   (b) $P(1 \leq W < 3)$.

The sample space S for the three tosses of the coin is:

SS	HHH	HHT	HTH	THH	THT	HTT	TTH	TTT
w	3	1	1	1	-1	-1	-1	-3

SS	HHH	HHT	HTH	THH	THT	HTT	TTH	TTT
w	3	1	1	1	-1	-1	-1	-3

Since the coin is biased, the probability of getting a head for one toss is 2/3 and the prob. of getting a tail is 1/3. We can get the probability mass function

$$
\begin{aligned}
P(x=-3) &= P(TTT) = \frac{1}{3}.\frac{1}{3}.\frac{1}{3} = \frac{1}{27} \\
P(x=-1) &= P(HTT) + P(THT) + P(TTH) = 3\left(\frac{2}{3}.\frac{1}{3}.\frac{1}{3}\right) = \frac{2}{9} \\
P(x=1) &= P(HHT) + P(HTH) + P(THH) = 3\left(\frac{2}{3}.\frac{2}{3}.\frac{1}{3}\right) = \frac{4}{9} \\
P(x=3) &= P(HHH) = \frac{2}{3}.\frac{2}{3}.\frac{2}{3} = \frac{8}{27}
\end{aligned}
$$

X	-3	-1	1	3
P(X)	1/27	2/9	4/9	8/27

Formula for the Mean of a Probability Distribution

The mean of a random variable with a discrete probability distribution is

$$\mu = X_1 P(X_1) + X_2 P(X_2) + \cdots + X_n P(X_n) = \sum X.P(X)$$

where X_1, X_2, X_3, $\cdots$, X_n are the outcomes and $P(X_1)$, $P(X_2)$, $P(X_3)$, $\ldots$, $P(X_n)$ are the corresponding probabilities.

Let X be a random variable with probability distribution f(x). The **mean, or expected** value, of X is

Let X be a random variable with probability distribution $f(x)$. The **mean**, or **expected value**, of X is

$$\mu = \sum_{x} x f(x) \qquad \textbf{(Discrete distribution)}$$

$$\mu = \int_{-\infty}^{\infty} x f(x) dx \qquad \textbf{(Continuous distribution)}$$

and the **variance** σ^2(sigma square) by

$$\sigma^2 = \sum_{x} (x-\mu)^2 f(x) \qquad \textbf{(Discrete distribution)}$$

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx \qquad \textbf{(Continuous distribution)}$$

Find the mean of the number of spots that appear when a die is tossed.

Since the sample space is 1, 2, 3, 4, 5, 6 and each outcome has a probability of , the distribution is as shown.

Outcomes X	1	2	3	4	5	6
P(X)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\mu = \sum X.P(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$
$$= \frac{1+2+3+4+5+6}{6} = \frac{\sum X}{N} = 3.5$$

If three coins are tossed, find the mean of the number of heads that occur.

The probability distribution is

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
	3C0	**3C1**	**3C2**	**3C3**

The mean is

$$\mu = \Sigma X \cdot P(X) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1\frac{1}{2} \text{ or } 1.5$$

Find mean of score per ball when a batsman score his fifty in 20 balls. Following this sequence of his scores

0, 1, 1, 2, 0, 4, 6, 6, 0, 4, 2, 1, 4, 0, 4, 2, 0, 6, 1, 6

$$\mu = \frac{50}{20} = 2.5$$

Make a probability distribution of score

Runs X	0	1	2	3	4	5	6
n_X	5	4	3	0	4	0	4
P(X)	$\frac{5}{20}$	$\frac{4}{20}$	$\frac{3}{20}$	0	$\frac{4}{20}$	0	$\frac{4}{6}$

$$\mu = \sum X \cdot P(X) = 0 \cdot \frac{5}{20} + 1 \cdot \frac{4}{20} + 2 \cdot \frac{3}{20} + 4 \cdot \frac{4}{20} + 6 \cdot \frac{4}{20}$$
$$= \frac{4}{20} + \frac{6}{20} + \frac{16}{20} + \frac{24}{20} = \boxed{\frac{50}{20} = 2.5}$$

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

- X denotes number of good components, $X = 0, 1, 2, 3$

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

- X denotes number of good components, $X = 0, 1, 2, 3$
- Total number of ways of selecting 3 components out of 7, $\boxed{C_3^7}$

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

- X denotes number of good components, $X = 0, 1, 2, 3$
- Total number of ways of selecting 3 components out of 7, $\boxed{C_3^7}$
- Selecting x good components out of 7 (4 good and 3 defective)

$$\boxed{f(x) = \frac{C_x^4 C_{x-3}^3}{C_3^7}}, \text{ where } x = 0, 1, 2, 3$$

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

- X denotes number of good components, $X = 0, 1, 2, 3$
- Total number of ways of selecting 3 components out of 7, $\boxed{C_3^7}$
- Selecting x good components out of 7 (4 good and 3 defective)

  $\boxed{f(x) = \dfrac{C_x^4 C_{x-3}^3}{C_3^7}}$, where $x = 0, 1, 2, 3$
- $\mu = \sum x f(x) = \frac{12}{7}$

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & \text{x} > 100; \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected life of this type of device.

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & x > 100; \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected life of this type of device.

$$E[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx \implies$$

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & x > 100; \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected life of this type of device.

$$E[X] = \mu = \int_{-\infty}^{\infty} xf(x)dx \implies E[X] = \int_{-\infty}^{100} xf(x)dx + \int_{100}^{\infty} xf(x)dx$$

$$E[X] = \int_{100}^{\infty} x \cdot \frac{20,000}{x^3} dx = 200$$

		PDF	Papulation
Mean μ	$E[X]$	$\sum_x x.f(x)$	$\frac{\sum X}{N}$
Variance	$E[(X-\mu)^2]$	$\sum_x (x-\mu)^2 f(x)$	$\frac{\sum (X-\mu)^2}{N}$
σ^2	$E[X^2]-\mu^2$ $= E[X^2]-(E[X])^2$	$\sum_x x^2.f(x)-\mu^2$	$\frac{\sum X^2}{N}-\mu^2$

Now let us consider a new random variable $g(X)$, which depends on X. For instance, $g(X)$ might be X^2 or $3X - 1$, and whenever X assumes the value 2, $g(X)$ assumes the value $g(2)$.

In particular, if X is a discrete random variable with probability distribution $f(x)$, for $x = -1, 0, 1, 2$, and $g(X) = X^2$, then the **mean**, or **expected value**, of $g(X)$, for following distribution, is

X	-1	0	1	2
g(X)	1	0	1	4
f(g(X))	f(-1)	f(0)	f(1)	f(2)

$$
\begin{aligned}
E[g(X)] = \mu_{g(x)} &= 0f(0) + 1[f(-1) + f(1)] + 4f(2) \\
&= (-1)^2 f(-1) + 0f(0) + (1)^2 f(1) + (2)^2 f(2) \\
&= \sum (X)^2 f(X) = \sum g(x) f(x)
\end{aligned}
$$

$$
\begin{aligned}
\mu_{g(x)} &= E[g(x)] = \sum_x g(x)f(x) && \textbf{(Discrete distribution)} \\
\mu_{g(x)} &= E[g(x)] = \int_{-\infty}^{\infty} g(x)f(x)dx && \textbf{(Continuous distribution)}
\end{aligned}
$$

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

X	4	5	6	7	8	9
f(X)	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

$$E\left[g(X)\right] = \sum g(X)f(X) = \sum \left(2X - 1\right) f\left(X\right)$$

$$(2(4)-1)(\frac{1}{12}) + (2(5)-1)(\frac{1}{12}) + (2(6)-1)(\frac{1}{4}) + (2(7)-1)(\frac{1}{4})$$

$$+ (2(8)-1)(\frac{1}{6}) + (2(9)-1)(\frac{1}{6}) = 12.67$$

Corollary

If a and b are constants, then

$$E[aX + b] = aE[X] + b$$
$$Var(aX + b) = a^2 Var(X)$$

X	4	5	6	7	8	9
f(X)	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E[2X - 1] = 2E[X] - 1 = 2[(4 + 5)/12 + (6 + 7)/4 + (8 + 9)/6] - 1$$
$$2[\frac{9}{12} + \frac{13}{4} + \frac{17}{6}] - 1 = 2\left(\frac{82}{12}\right) - 1 = 12.67$$
$$Var(2x - 1) = ?$$

A box contains 5 balls. Two are numbered 3, one is numbered 4, and two are numbered 5. The balls are mixed and one is selected at random. After a ball is selected, its number is recorded. Then it is replaced. If the experiment is repeated many times, find the variance and standard deviation of the numbers on the balls.

Number on balls X	3	4	5
Probabilites f(X)	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$

$$\mu = E[X] = \sum Xf(X) =$$

$$\sigma = E[X^2] - \mu^2 = \sum X^2 f(X) - \mu^2$$

One thousand tickets are sold at \$ 1 each for a color television valued at \$350. What is the expected value of the gain if you purchase one ticket?

	Win	Lose
Gain X	\$349	$-$1
Probability $P(X)$	$\frac{1}{1000}$	$\frac{999}{1000}$

$$E(x) = \$349.\frac{1}{1000} + (-\$1)\frac{999}{1000} = -\$0.65$$

The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density $f(X) = \begin{cases} 2(X-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

Find the mean and variance of X

$$\sigma^2 = E[(X-\mu)^2] = E[X^2] - \mu^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

Note That $E[X^2] \neq (E[X])^2$

Calculate the variance of $g(X) = 2X + 3$, where X is a random variable with probability distribution

X	0	1	2	3
F(X)	1/4	1/8	1/2	1/8

$$Var[g(X)] = Var[2X + 3] = ?$$

If $g(X) = X^r$ for $r = 0, 1, 2, 3, \cdots$, an expected value called the r^{th} moment about the origin of the random variable X, which we denote by μ_r' .

$$\mu_r' = E[X^r] = \sum_x x^r f(x) \qquad \textbf{(Discrete distribution)}$$

$$\mu_r' = E[X^r] = \int_{-\infty}^{\infty} x^r f(x) dx \qquad \textbf{(Continuous distribution)}$$

Note that

$$\mu = \mu_1', \qquad \sigma^2 = \mu_2' - \left(\mu_1'\right)^2$$

Suppose that an antique jewelry dealer is interested in purchasing a gold necklace for which the probabilities are 0.22, 0.36, 0.28, and 0.14, respectively, that she will be able to sell it for a profit of \$250, sell it for a profit of \$150, break even, or sell it for a loss of \$150. What is her expected profit?

4.8 Let $X =$ profit. Then

$$\mu = E(X) = (250)(0.22) + (150)(0.36) + (0)(0.28) + (-150)(0.14) = \$88.$$

1. $f(x) \geq 0$
2. $\sum_x f(x) = 1$
3. $P(X = x) = f(x)$
4. $F(x) = \sum_{x_j \leq x} f(x_j)$
5. $\mu = \sum_x xf(x)$
6. $\sigma^2 = \sum_x (x - \mu)^2 f(x)$
7. $\mu_{g(x)} = E[g(x)] = \sum_x g(x)f(x)$
8. $\mu'_r = E[X^r] = \sum_x x^r f(x)$

1. $f(x) \geq 0$, for all $x \in \mathbf{R}$.
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_a^b f(x)dx$.
4. $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$
5. $\mu = \int_{-\infty}^{\infty} xf(x)dx$
6. $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$
7. $\mu_{g(x)} = E[g(x)] = \int_{-\infty}^{\infty} g(x)f(x)dx$
8. $\mu'_r = E[X^r] = \int_{-\infty}^{\infty} x^r f(x)dx$

1. The probability distribution of the discrete random variable X is

$$f(x) = C_x^3 \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3.$$

   Find the mean of X, also Find the expected value of the random variable $g(X) = X^2$.

2. The density function of the continuous random variable X, the total number of hours, in units of 100 hours, that a family runs a vacuum cleaner over a period of one year, is given in

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

   Find the average number of hours per year that families run their vacuum cleaners.

4.13 $E(X) = \int_0^1 x^2\, dx + \int_1^2 x(2-x)\, dx = 1$. Therefore, the average number of hours per year is $(1)(100) = 100$ hours.

QUIZ#3

1. In a gambling game, a woman is paid \$3 if she draws a jack or a queen and \$5 if she draws a king or an ace from an ordinary deck of 52 playing cards. If she draws any other card, she loses. How much should she pay to play if the game is fair?
2. A private pilot wishes to insure his airplane for \$200,000. The insurance company estimates that a total loss will occur with probability 0.002, a 50% loss with probability 0.01, and a 25% loss with probability 0.1. Ignoring all other partial losses, what premium should the insurance company charge each year to realize an average profit of \$500?